

PROJECT REPORT

On

S.C.O.U.T. (Spatial Calculation & Obstacle Understanding Terrain-vehicle)



Robotics & Drone Lab [22MEC 37N]

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ABSTRACT

This project presents the design and development of S.C.O.U.T. (Spatial Calculation & Obstacle Understanding Terrain-vehicle), a 4-wheel-drive autonomous obstacle-avoidance robot. The primary objective is to build a reactive, intelligent navigation system capable of safely traversing unknown environments without human intervention. The system utilizes an Arduino Uno microcontroller coupled with an L298N motor driver for mobility. For spatial awareness, we mounted an HC-SR04 Ultrasonic Sensor on a 180-degree servo motor.

Crucially, the system uses high-frequency sound waves (ultrasonic echolocation) instead of optical infrared sensors, ensuring reliable operation regardless of ambient lighting conditions like bright sunlight. The robot's algorithm continuously measures forward distance; upon detecting a barrier within 50 cm, it halts, reverses, and sweeps its sensor left and right. It then mathematically compares these distances to determine the optimal escape route. Experimental results over 100 trials demonstrated a 95% detection success rate and 88% turn accuracy, proving the system is a highly robust foundational platform for autonomous driving studies.

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CHAPTER 1 INTRODUCTION

1.1 Background of the problem

Autonomous navigation is the core of modern robotics, from robotic vacuum cleaners to planetary rovers. A fundamental requirement for any autonomous vehicle is the ability to perceive its environment and dynamically react to sudden obstacles to prevent collisions and system damage.

1.2 Problem Statement

Many basic robotic systems use static, forward-facing sensors that lack peripheral spatial awareness. When confronted with an obstacle, these robots only know to stop; they lack the intelligence to "look around" and determine the most efficient alternative path, frequently resulting in getting trapped in corners.

1.3 Objectives of the project

- To design and assemble a robust 4-wheel-drive mobile robotic chassis.
- To implement a dynamic "look-around" scanning mechanism using a servo-mounted distance sensor.
- To program an intelligent algorithm that compares spatial variables (left distance vs. right distance) to make logical routing decisions.

1.4 Overview of project relevance

The logic and sensor integration developed in this project form the foundational architecture for more complex autonomous systems, heavily mirroring the rudimentary decision-making loops found in self-driving cars and automated guided vehicles (AGVs) in industrial warehouses.

CHAPTER 2 LITERATURE SURVEY

2.1 Summary of previous work or related research

Early iterations of obstacle-avoidance robots frequently relied on simple mechanical bump sensors (limit switches) which required physical contact to trigger. Later models transitioned to basic optical sensors, making the systems purely reactive but non-predictive.

2.2 Existing technologies and solutions (The IR Flaw)

Currently, Infrared (IR) Obstacle Avoidance Sensors are widely used in introductory robotics. IR sensors work by emitting a beam of infrared light and waiting for it to reflect off an object back into a photodiode receiver.

- **The Critical Limitation:** The sun is a massive source of natural infrared radiation. When IR-equipped robots operate outdoors or near bright windows, the ambient infrared light floods the receiver. The sensor becomes "blinded," unable to distinguish between its own reflected light and the sunlight, causing the robot to either freeze permanently or crash blindly into walls.

2.3 Comparative analysis with your project and Research Gaps

To overcome the glaring limitations of optical IR technology, our project explicitly utilizes an Ultrasonic Sensor (HC-SR04). Instead of light, ultrasonic sensors emit high-frequency sound waves. Because sound waves are completely unaffected by ambient lighting, S.C.O.U.T. functions flawlessly in glaring sunlight, complete darkness, and heavily dynamic indoor environments. Furthermore, by mounting the sensor on a servo, our system addresses the research gap of "tunnel vision" in standard robots, allowing a 180-degree sweep before committing to an evasive maneuver.

CHAPTER 3 METHODOLOGY

3.1 SYSTEM DESIGN

Hardware Requirements

- Arduino Uno R3 (Microcontroller)
- L298N Motor Driver (for PWM speed control)
- HC-SR04 Ultrasonic Sensor (Distance calculation)
- Micro Servo Motor (SG90) (For 180-degree sweeps)
- 4x DC Gear Motors and Wheels (4WD Chassis)
- 2x 18650 Lithium-Ion Batteries

Software Requirements

- Arduino IDE (C++ programming environment)

Block Diagram & Circuit Diagram

(Insert Block Diagram & Circuit Diagram Images Here showing the Arduino connected to the L298N, Servo, and Ultrasonic sensor)

Workflow / Algorithm

- Initialize motors, center the servo (90°), and begin calculating forward distance.
- If distance > 50 cm: Move Forward continuously.
- If distance < 50 cm:
 - Halt the robot and reverse for 800ms to create pivoting clearance.
 - Sweep servo to 160° (Left) and record distance.
 - Sweep servo to 20° (Right) and record distance.
 - Compare values: If Left Distance > Right Distance, execute a zero-point turn Left. Otherwise, turn Right.

3.2 IMPLEMENTATION

Hardware Assembly & Sensor Calibration

The L298N motor driver is powered directly by the 18650 battery pack to ensure the high current draw of four DC motors does not starve the Arduino. The ultrasonic sensor relies on the speed of sound (340m/s). We calibrated the distance output using the formula: $\text{Distance} = (\text{Duration} \times 0.034) / 2$.

Challenges Faced

System latency was the primary challenge. If the robot moved too quickly, it closed the distance to the obstacle before the sound wave could complete its travel and trigger the software interrupt. We resolved this by precisely capping the PWM speedVal variable to 110, finding the optimal balance between physical momentum and micro-processing time.

Full Arduino Code

```
#include <SoftwareSerial.h>
#include <Servo.h>

#define ENA 5          // PWM left motor
#define IN1 6
#define IN2 4

#define ENB 3          // PWM right motor
#define IN3 8
#define IN4 7

#define TRIG A0
#define ECHO A1

#define SERVO_PIN 9

Servo scanner;

int speedVal = 110;

void setup() {
  pinMode(ENA, OUTPUT);
  pinMode(ENB, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);

  scanner.attach(SERVO_PIN);
  scanner.write(90); // center
  Serial.begin(9600);
}

void loop() {
  int frontDist = getDistance();
  Serial.print("Front: ");
  Serial.println(frontDist);

  if (frontDist > 50) {
    forward();
  }
  else {
    stopRobot();
    delay(300);

    // Move back ~10 cm
    backward();
    delay(800);
    stopRobot();
    delay(300);

    // Check Left
    scanner.write(160);
    delay(600);
    int leftDist = getDistance();
```

```

    // Check Right
    scanner.write(20);
    delay(600);
    int rightDist = getDistance();

    // Center servo
    scanner.write(90);
    delay(400);

    if (leftDist > rightDist) {
        turnLeft();
        delay(1000);
    } else {
        turnRight();
        delay(1000);
    }
    stopRobot();
    delay(200);
}

int getDistance() {
    digitalWrite(TRIG, LOW);
    delayMicroseconds(2);
    digitalWrite(TRIG, HIGH);
    delayMicroseconds(10);
    digitalWrite(TRIG, LOW);

    long duration = pulseIn(ECHO, HIGH, 30000);
    int distance = duration * 0.034 / 2;
    if (distance == 0) distance = 300;
    return distance;
}

void forward() {
    analogWrite(ENA, speedVal); analogWrite(ENB, speedVal);
    digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
    digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
}

void backward() {
    analogWrite(ENA, speedVal); analogWrite(ENB, speedVal);
    digitalWrite(IN1, LOW); digitalWrite(IN2, HIGH);
    digitalWrite(IN3, LOW); digitalWrite(IN4, HIGH);
}

void turnLeft() {
    analogWrite(ENA, speedVal); analogWrite(ENB, speedVal);
    digitalWrite(IN1, LOW); digitalWrite(IN2, HIGH);
    digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
}

void turnRight() {
    analogWrite(ENA, speedVal); analogWrite(ENB, speedVal);
    digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
    digitalWrite(IN3, LOW); digitalWrite(IN4, HIGH);
}

void stopRobot() {

```

```
digitalWrite(IN1, LOW); digitalWrite(IN2, LOW);  
digitalWrite(IN3, LOW); digitalWrite(IN4, LOW);  
}
```

CHAPTER 4 RESULTS & DISCUSSION

4.1 Output Readings, Graphs, Tables

To validate the efficiency of S.C.O.U.T., we subjected the robot to 100 trials spread across 4 distinct environmental cases (25 trials per case) to calculate Detection Success, Turn Accuracy, and Sensor Precision.

Table 4.1: Experimental Trial Data (25 Trials per Case)

Test Case (Obstacle Type / Environment)	Total Trials	Detection Success	Turn Accuracy	Sensor Precision
Case 1: Solid Wall (Smooth Floor)	25	25	24	25
Case 2: Dark Object (Smooth Floor)	25	25	23	24
Case 3: Soft Material (Carpeted Floor)	25	21	20	20
Case 4: Narrow Object (Chair Leg)	25	24	21	23
TOTALS	100	95	88	92

Performance Metrics

- Detection Success: 95% (95/100)
- Turn Accuracy: 88% (88/100)
- Sensor Precision: 92% (92/100)

4.2 Observations and Performance Analysis

- **Environmental Reliability:** As predicted, the ultrasonic sensor performed with equal accuracy against solid walls and dark objects (100% detection success in Cases 1 and 2), effectively bypassing the limitations of light-sensitive sensors.
- **Material Sensitivity:** The 5% detection failure rate occurred primarily in Case 3. Soft, highly porous materials (like thick blankets or carpeted walls) absorb ultrasonic sound waves rather than reflecting them, resulting in occasional delayed detection.

- **Turn Accuracy Drop:** Turn accuracy dropped to 88% largely due to floor friction. On carpeted surfaces, the increased rolling resistance reduced motor speed responsiveness, occasionally causing the robot to slightly under-rotate during its calculated escape maneuver.

CHAPTER 5 CONCLUSION

5.1 Conclusions Drawn

The project successfully achieves its objective of creating an autonomous, environmentally aware robotic system. By integrating a "Sense-Plan-Act" logic loop with a 180-degree sweeping ultrasonic sensor, S.C.O.U.T. vastly surpasses the capabilities of basic reactive machines. The explicit engineering choice to utilize acoustic waves over optical/IR sensors resulted in a robust system completely immune to ambient light interference.

5.2 Future Enhancements

Future iterations of S.C.O.U.T. could be drastically improved by integrating rotary wheel encoders. Encoders would allow the microcontroller to measure exactly how many rotations the wheels have made, transitioning the system from "time-based" turning (`delay(1000)`) to precise "degree-based" turning. Furthermore, upgrading to a 360-degree LiDAR sensor would allow the robot to generate real-time 2D maps of its environment for advanced pathfinding.

REFERENCES

- [1] C. Bell, Beginning Sensor Networks with XBee, Raspberry Pi, and Arduino, Apress, 2020.
- [2] Arduino Documentation, "Ultrasonic Sensor HC-SR04 Interfacing," arduino.cc. [Online]. Available: <https://docs.arduino.cc/>.
- [3] STMicroelectronics, "L298N Dual Full-Bridge Driver Technical Specifications Datasheet," st.com.
- [4] J. Borenstein and Y. Koren, "Real-time obstacle avoidance for fast mobile robots," IEEE Transactions on Systems, Man, and Cybernetics, vol. 19, no. 5, pp. 1179-1187, Sept.-Oct. 1989.